Beyond Concept Detection: The Potential of User Intent for Image Retrieval

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ABSTRACT

Behind each photographic act is a rationale that impacts the visual appearance of the resulting photo. Better understanding of this rationale has great potential to support image retrieval systems in serving user needs. However, at present, surprisingly little is known about the connection between *what* a picture shows (the literally depicted conceptual content) and *why* that picture was taken (the photographer intent). In this paper, we investigate photographer intent in a large Flickr data set. First, an expert annotator carries out a large number of iterative intent judgments to create a taxonomy of intent classes. Next, analysis of the distribution of concepts and intent classes reveals patterns of independence both at a global and user level. Finally, we report the results of experiments showing that a deep neural network classifier is capable of learning to differentiate between these intent classes, and that these classes support the diversification of image search results.

CCS CONCEPTS

• **Information systems → Multimedia information systems**; **Information retrieval**; **Information retrieval diversity**; **Image search**; *Information systems applications*; *Query intent*; *Personalization*; *Rank aggregation*;

KEYWORDS

Multimedia retrieval; user intent; multimedia indexing; diversification;

1 INTRODUCTION

Today's challenges in the area of multimedia content analysis and retrieval have attracted enormous attention from both academia and industry. By employing techniques for processing and analyzing multimedia content, we have successfully brought automatic recognition of the literally depicted content of images (e.g., object recognition, scene labeling) to the next level [5].

However, because researchers have focused nearly exclusively on the literally depicted content (the "what" dimension), the research field of multimedia content analysis have overlooked opportunities to better understand the reasons for which people take photos in the first place (the "why" dimensions).

Figure 1: Example search results for the query "sailing". The intent (goal) of the photographer is different for each picture. Plausible characterizations of these goals are: provide an overall impression of the space in the wider world where sailing takes place (top left), depict a sailboat as an object (top right), capture a portrait of someone sailing (bottom left), capture information on sailing from another media source (bottom right).

1.1 The Potential of Intent

The area of multimedia information retrieval has seen exceptions to this tendency towards "what" at the cost of "why", which include recent work on user search intent [11, 15, 18]. The purpose of this paper is to build on previous work and to provide concrete, large-scale substantiation of the importance of moving multimedia analysis beyond concept detection, and the potential of techniques that automatically detect visual classes related to user intent to improve image retrieval. Contrasts in user intent are illustrated in Fig. 1, which shows four images uploaded to Flickr relevant to "sailing". Despite the fact that all four are relevant to the same topic, clear differences can be observed.

We argue that user intent should be exploited in image retrieval since the goals of the photographer provides a simple, easily understandable explanation for the differences observed between photos. If we were to restrict the descriptions of photos used for indexing to containing only visual concepts, then both pictures in the top row could be described as "Photo depicting a sailboat in water". We

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could then capture the fact that users see a clear difference between the two by describing the first as "sailboat in a lot of water" and "sailboat near a dock". However, such descriptions are not entirely satisfying since the exact amount of water or the presence of the dock is not the key characteristic differentiating these photos. Instead, the difference can be simply and naturally explained by user intent, i.e., the goal of the photographer in capturing the image.

We build on the assumption that user intent, i.e., the goals that users are pursuing when they take photos or search for images, has visual reflexes that can be captured by automatic visual classifiers. Our motivation for assuming a connection between the reason why a user takes a photo, and the visual content of the photo, is the phenomenon of *intentional framing*. Riegler et al. [18] define intentional framing as, 'the sum of the choices made by photographers on exactly how to portray the subject matter that they have decided to photograph.' This definition implies that intentional framing is observable in the photo, and also captured the fact that intent cannot be reduced to the concepts that are literally depicted in the image. Our notion of intent is related to the idea of *broad latent aspects* from conventional web search [24]. This type of query aspect is described by [24] as having two properties: broad latent aspects apply to a broad set of queries, and, users queries frequently leave these aspects unspecified.

Note that the user intent behind a query does not reduce to a sub-topic of a query. To illustrate this point, we return to Fig. 1 to discuss the two pictures in the bottom row. If we considered these pictures sub-topics of sailing, then we would loose connection with the intent of capturing a portrait and the intent of capturing media information that apply to a large range of queries beyond "sailing". In other words, intent cross-cuts the topic (conceptual content) of a photo and provides us with an additional, easily understandable dimension with respect to which images can be indexed for image retrieval.

1.2 Challenges and Contributions

In this work, we investigate the ability of automatic intent inference to improve image retrieval, with a focus on retrieval algorithms that diversify image results lists. Given that intentional framing is noticeable to users as visible in photos, but cannot be directly reduced to topical or conceptual categories, it is surprising that user intent has yet to be fully exploited in image retrieval systems. Because the intent of the user is actually the goal of the user, we anticipate that user intent potentially has an important contribution to make to image retrieval. We believe that there are three major roadblocks that explain the relative lack of research on intentional framing in image search, which we now discuss in turn. The three major contributions of this paper address each of these three roadblocks.

The first roadblock is the lack of intent taxonomies (definitions of intent classes) and data sets annotated with intent labels. With the exception of early work by Lux et al. [15], we do not know of work that has asked photographers about their intent. However, asking photographers might not always be necessary. As mentioned above, people also perceive intent classes when they look at images. However, there is no standard set of intent classes that has universal applicability, and inner-annotator agreement on intent judgments is difficult to achieve. We address this roadblock by moving away from the idea that we should assume that a single authoritative set of intent classes is necessary before progress can be made on intent-based image retrieval. Instead, we pursue the idea that any well-informed intent taxonomy that has been applied consistently in order to label an image data set will be able to drive forward the state of the art in applying user-intent to improve image retrieval. To this end, we build an intent taxonomy on the basis of the analysis of a large collection of social images by a single expert annotator. The first contribution of this paper is this taxonomy and a large set of images labeled with the intent classes that it contains. This resource will allow reproduction of our research results and serves as a point of reference for future extension or modification of the intent taxonomy.

The second roadblock standing in the way of applying intent-based approaches to image retrieval is general assumptions about the nature of intent. We believe that the currently dominant assumption is that the visual variability among photos associated with a single intent class is too high for a classifier to be able to generalize. Likewise, the visual similarity among photos associated with different intent classes is also assumed to be too high. The second contribution of this paper is to show that these assumptions should not be made: with enough labeled data it is possible to train a classifier capable of automatically inferring the intent classes of images.

The third roadblock is the assumption that the usefulness of intent in image retrieval is likely to be limited. The third contribution of this paper is a set of experiments on multimodal social image retrieval that show that our intent classifier is able to contribute productively to the diversification of image search results.

The remainder of the paper is organized as follows. In Section 2, we cover related work, focusing on multimedia intent taxonomies that have been proposed in the literature. Then, in Section 3 we describe the iterative process by which we create our intent taxonomy and label our image data set. Next, we train a classifier using the labeled data, and report the results of experiments demonstrating its effectiveness. Finally, in Section 6, we apply our intent classes to information retrieval. Section 7 of the papers provides conclusions and an outlook.

2 RELATED WORK

This section covers research that has trained image classifiers to predict intent, as well as papers that have proposed multimedia intent taxonomies. We use this work to inform the creation of our own taxonomy on the basis of a large collection of social photos, which is discussed in Section 3.

2.1 Intent in Image Search

In [11], a survey of user intent in multimedia search, an overview is provided of papers that have made use of user intent to improve image search. Note that we are interested in papers that consider "user intent" to be the goal of the user, and not in work that users "intent" as a synonym for sub-concept, sub-topic, or query aspect. An early effort [12] used a a content-based intent classifier to adapt the display of search results. Recently, [20] used content-based image classification as one of a range of session-derived signals used to study user intent during image search. This work is interesting

because it focused on predicting intent by processing the sequence of images that users view in an image search session.

2.2 Taxonomies of Image Intent

The earliest taxonomy of image search intent is most probably that of Fidel et al. [4], who define intent along a spectrum from a *Data Pole* (obtain source of information) and the *Information Pole* (obtain an example of something or and object). To create their taxonomy they first categorized images into 12 attribute classes based on Jorgensen's research [10]. A user study was then conducted on 100 image search requests.

Other early work includes Cunningham et al. [2], who investigate 98 video search queries and categorize the user search intent into 8 groups. For example, *mental state* explicit the reference to subject's emotional state, *visual* explicit references to visual aspects to target video, *MSM* explicit references to mainstream media.

Very valuable image intent taxonomies have bee proposed on the basis of use studies. Lux et al. [13] conducted user research on 20 image search users on Flickr, and build an intent taxonomy with four classes: knowledge orientation, transaction, mental image and navigation.

By interviewing 10 people covering 40 different situations, Lux et al. [14, 15] studied the user intentions on photo production. The result was a intent taxonomy containing six reasons why users take pictures: *preserve a good feeling*, *preserve a bad feeling*, *share with family and friends*, *share with public*, *support a task* and *recall a specific situation*.

The work closest to our own is Hanjalic et al. [6], who employed a social-web mining approach, analyzing 4512 questions posted on Yahoo Answers that contained certain terms (i.e. "find" and "video"). The questions were sorted by crowdsourincg workers and then, using a card sorting process similar to our own manual content analysis process, and five classes were defined: Information, Experience:Learning, Experience: Exposure, Affect, and Object. Instead of investigating video, we investigate images. The commonality is that our work also exploits a large collection of social media to identify a set of intent classes that may not be absolutely universal, but is "good enough" to support tasks such as retrieval.

3 INTENT DISCOVERY

In this section, we describe the intent taxonomy (set of intent classes). The taxonomy is created on the basis of sub-set of a large-collection of social images, namely, the YFCC100M data set [21]. We create the classes using a manual content analysis [16] analysis approach. The intent classes are then used to label the data set. We are interested in demonstrating the ability of intent to go beyond concept detection. For this reason, we need to ensure that the data set contains images depicting a wide range of concepts, but also that there are enough images per concept. This section first discusses the specifics of the data set creation, and then describes the content-analysis procedure which was used to create the intent taxonomy.

3.1 Data set and set up

The first step in creating the data set is to define a list of concepts $C = \{c_1, c_2, ..., c_k\}$ that we would like to focus on. We chose the NUS-Wide concepts for several reason: this concept set is comparable to other concept sets used in the literature, it corresponds to the most frequent tags in Flickr, it includes both general and specific concepts, and finally, the concepts are of different types, including scene, object, event, program, people and graphics [1].

The concepts are used as queries to retrieve images from the YFCC100M data set with a tag-based retrieval system. We choose YFCC100M for a variety of reasons: 1. To our knowledge it is the largest public image collection that has ever been released. 2. It offers social images associated with user information. 3. It is publicly available, which supports reproducibility. Flickr has a batch tagging function that allows users to assign tags to a set of uploaded images [23]. Since batch tagging might affect the robustness of our intent data set, we only retain a maximum of three images with the same tags. For each concept (query), we collect top-200 relevant images. If less than 200 images are returned by the query, we use the total number of images returned. Our final data set contains 15618 images.

3.2 Taxonomy Creation

Next we turn to describing the content analysis approach by which we created the intent taxonomy. The approach involves iterative labeling by an expert annotator. The expert acquired his expertise by reading the papers in Section 2, and studying the taxonomies proposed there. During the labeling process he allowed his understanding of intent to evolve guided by what he observed in the data.

Figure 2: Example of two photos that appear similar at first glance, but ultimately land in two different intent categories, because of the difference in photographer goal.

Specifically, our manual content analysis approach proceeds by examining images in turn, and applying a preliminary intent label. Each new image is judged as either belonging to an existing class, or potentially requiring the introduction of a new class. Before introducing a new class, the annotator returns to the previous annotated images to ensure that it is not possible to accommodate the new image by updating the description of an existing class. If no existing class can be extended to accommodate the new image, a new intent class is introduced. Note that the process requires multiple iteration over the whole data set, and represents a full person month of image annotation work.

The two images in Figure 2 demonstrate the process with a hypothetical example. The left image is previously labeled with

"portrait" and the right image is new. The annotator examines all previous images and realizes that the right image does not fit into the "portrait" class because the picture was not take with the same goal as a portrait. Another class, *situation_documentation* is then added to the taxonomy. The final taxonomy is shown in Table 1.

3.3 Validation of the Intent Taxonomy

For the purpose of validating that intent class must be considered independently of topics, an analysis of our intent taxonomy was carried out from two perspectives: how intent classes i were shared among concept groups c and how concepts c are shared within different intent classes i. We discuss each here in turn.

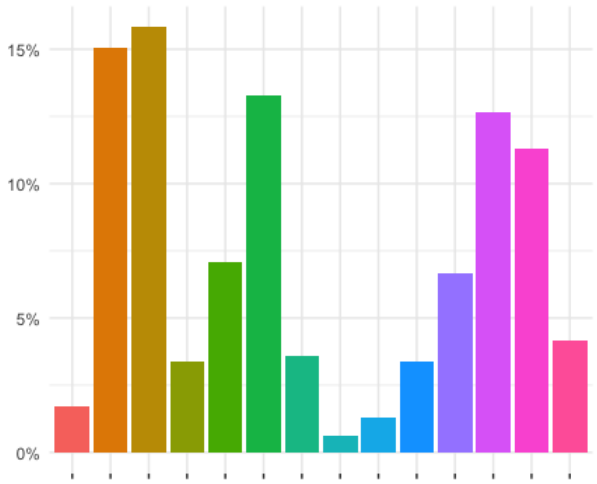

Figure 3: Distribution of intent classes, from left to right: art, candid, landscape, macro, media capture, portrait, product presentation, product presentation by person, situation documentation, social event private, social event public, setting, structure, wildlife.

In general, as can be seen in Figure 3, the 81 concept categories that form the basis of our data set are distributed across various intent classes. For example, the intent classes *portrait*, *candid*, *setting*, *media_capture* and *landscape* are found associated with images from 67 out of the 81 concepts. On the other hand, some intent classes did not show such generality. The intent classes *product_presentation_by_person*, *domesticate* and *situation_documentation* are found to be associated with 8–16 concepts. In average, an intent class is associated with **46.21** different concepts. This means that each intent class is associated on average with just over half of all concepts (to be exact **56.79%** of all concepts, with a standard deviation of **23.95**).

We turn now to look at how intent classes are distributed across concept categories. On average a concept category is associated with just over 8 intent classes (to be exact: **8.42** and **2.01** standard deviation.) These observations validate our intent taxonomy in the sense that they show that intent classes cross-cut image topic as represented by topic category, and that topics are associated with multiple intents. We note that the fact that our intent classes are effective for image search result diversification, provides further validation for our taxonomy. These experiments will be discussed in Section 6.

4 DISTRIBUTION OF INTENT CLASSES

In order to explore the distribution of intent classes over concepts, we ranked the concepts with respect to the number of intent classes with which they are associated. The top-10 concepts are listed in Table 2 and the bottom-10 in Table 3.

Concept c	Num Intents	Entropy
sun	12	3.16
frost	12	2.06
fish	12	3.09
tiger	11	2.80
animal	11	1.57
boat	11	2.61
earthquake	11	2.96
window	11	2.90
forest	11	2.86
fox	11	3.18

Table 2: Top-10 concepts ranked by the number of intent classes associated with the concept in the image collection.

Concept c	Num Intents	Entropy
coral	2	0.72
running	5	2.07
dog	5	1.52
elk	6	2.01
dancing	6	2.10
statue	6	1.83
plane	6	1.81
buildings	6	2.28
whale	6	1.67
surfing	6	1.93

Table 3: Bottom-10 concepts ranked by the number of intent classes associated with the concept in the image collection.

We observe that concepts differ with respect to the number and distribution of associated intent classes. For some concepts, there are photos in the collection associated with a wide range of different intent classes. For other concepts, the the number of intent classes is more limited. We capture these patterns with the entropy, which is reported in the tables.

More insight is provided by looking at four typical concepts in greater depth: *build*, *building*, *fox* and *wedding* as illustration in Figure 4. We see that concepts are directly associated with one dominant intent class, such as *bird* (top left) and *buildings* (top right). Other concepts are associated with multiple dominant intents. For example, for *wedding*, **78%** of the photos were captured related to people (*portrait* and *candid*) or aims at record a private social event. We noticed that for some concepts, such as *fox*, intent classes corresponded closely to what would be more conventionally

Class	Description
product_presentation	photos made to demonstrate or sell an item or a product.
product_presentation_by_person	photos made to demonstrate or sell an item or a product worn by or held by a human model.
social_event_public	photos made to record social events held for the public.
social_event_private	photos made to record planned social events held for a certain group of people.
situation_documentation	photo made to document desired or undesired situation.
landscape	photos made to shows spaces within the world, sometimes vast and unending.
macro	photos made to show extreme close-up photos of very small subjects too small to be noticed.
structures	photos made to demonstrate landmarks, buildings and similar structures.
setting	photos made to depict a inanimate objects, could be either nature or man-made, reflect specific aspect of the object.
portrait	photo of a person or a group of people that captures the characteristics of the subject(s)(who are aware they are being photographed).
candid	photo of a person or a group of people that capture the characteristics of the subject(s)(who are not aware they are being photographed).
wildlife	photos capture varies forms of wildlife in their nature habitat.
media_capture	image captured to carry out other information source.
art	photos captured to demonstrate abstract, creative vision of a photographer.

Table 1: Taxonomy of intent classes resulting from our intent discovery process

considered different interpretations of an ambiguous query. In a social media context, "fox" it might be an animal, a logo, or even a band of musicians, as reflected in our data set.

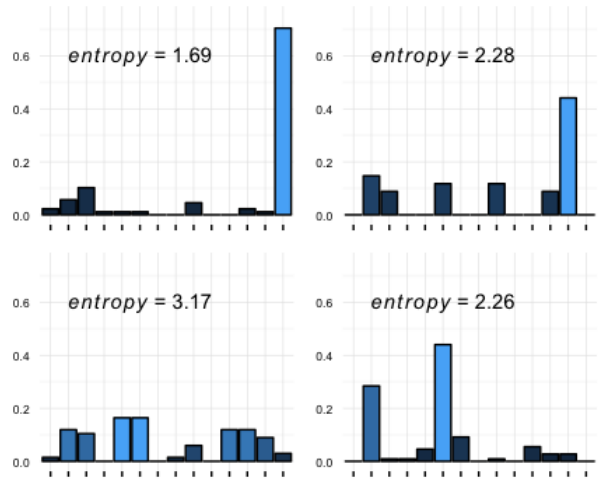

Figure 4: Distribution of intent classes with respect to four selected concepts plus entropy: bird (1.69), building (2.28), fox (3.17), wedding (2.26). Intent classes from left to right: art, candid, landscape, macro, media capture, portrait, product presentation, product presentation by person, setting, situation documentation, social event private, social event public, structure, wildlife.

To create the ground truth, we analyzed the distribution of intent classes over users. We aggregated the data set by user id, and filtered out users who had captured less than five photos, resulting in a set of 221 users. Figure 5 shows the entropy over the 14 intent classes with respect to each user.

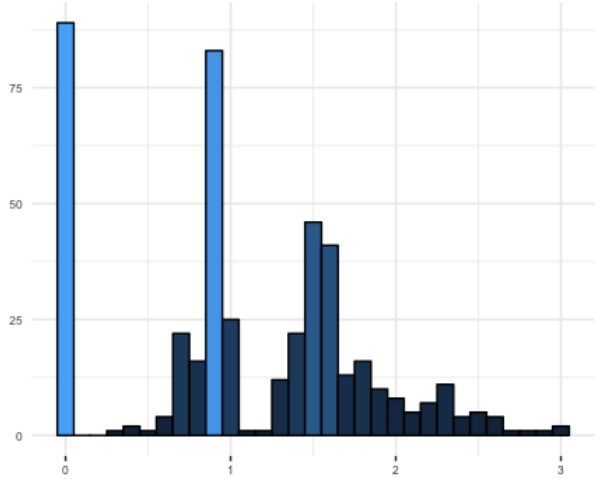

Figure 5: Intent entropy distributed over users, x-axis indicates entropy value, y-axis indicates number of users.

We see that the images of some users are associated with a stable set of intent classes and some have are unevenly associated with intent classes. A large number of users have one particular intent (i.e., entropy is 0). We note that this observation might be an artifact of the fact that for quite a few users the number of images per user is quite small. Overall the patterns in Figure 5 suggests that individual users have individual patterns of intent. We do not build further on this insight here, but only point out that in the future user-specific intent patterns could also be relevant for personalizing image search.

5 INTENT CLASSIFICATION

Next, we turn to investigate whether a classifier can learn to predict the intent class of an image. To this end, we adopt a conventional transfer learning scheme. Transfer learning consists of using models that were trained for a certain task and leveraging the knowledge that they acquired on a different, but related task [17]. In our case, VGGNet [19] was adopted to extract visual features from image pixels (originally trained on ImageNet [3]). The last fully connected layer (going from 2048 neurons to 1000 class scores) was removed and the rest of the networks serves as a feature extractor. We re-trained a Softmax classifier using a cross-entropy Softmax loss on our image data set annotated with 14 intent classes using 70% of the intent data set. Meanwhile, 25% of the images were held out for the validation purpose and we left 5% for future analysis. Before training, all images were re-sized into (224,224) and we applied random horizontal flipping, chopping and re-scaling for data augmentation. We evaluate with respect to accuracy.

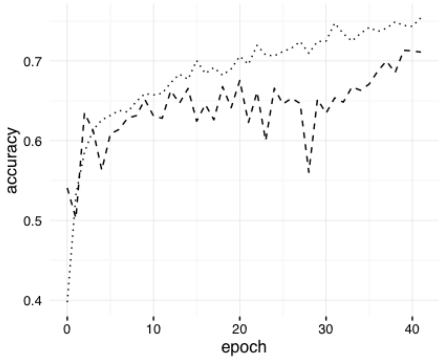

Figure 6: Intent class prediction: training set accuracy (dotted line) and validation set accuracy (dashed line).

As shown in Figure 6, our model achieved 71% accuracy on the validation set. This performance level demonstrates that there is enough visual stability within intent classes to allow a classifier to generalize them. We expect with a larger labeled intent data set, model performance could be further improved.

5.1 Discriminative Ability of Intent Classes

Next we analyze the ability of the classifier to distinguish intent classes in more depth. The data set we use here is the 5% of the labeled data (ca. 800 images) that was reserved (as mentioned above) for future analysis. Employing our trained model for intent classification, we predicted the outcomes for this small proportion of data.

As reflected in Figure 8, for most of the intent classes, over 70% of the images in the data set belonging to that category can be correctly predicted. However, for some particular intent classes, it is still non-trivial for our classifier to capture the patterns of the images belonging to this class. For example, for the classes *art* and *product_presentation*, our intent classifier can only identify 31.25% and 37.5% images correctly. We attribute this observation to the visual diversity of the images in these intent classes. However, we also point out, as Figure 8 indicates that for these two intent classes, there are fewer images in our data set compared to other classes.

Figure 7: Intent predicted as *landscape* from four different concepts, *road, zebra, city space, reflection.*

It is also interesting to observe that certain intent classes are easily confused. This effect occurs case of *candid* and *portrait*. By definition, we differentiate *candid* and *portrait* by the awareness of person or people appearing in the photo. Awareness is reflected in a subtle difference in direction of the gaze of the person in the image, and people judging images also might confuse these classes. Moreover, there could be multiple persons within an image (some aware while others not).

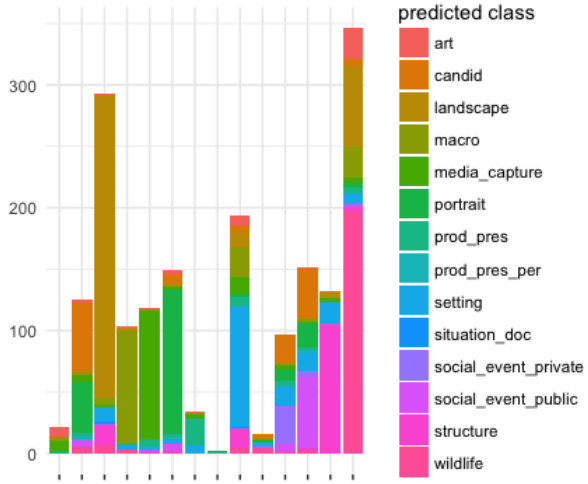

Figure 8: Discriminative ability of intent classes: bars are the intent classes (alphabetical order left to right). Height of the bars indicates the number of images in each class. Colors indicate class confusion.

6 SEARCH RESULT DIVERSIFICATION

An effective image retrieval system should be able to present results that are both relevant and that are covering different (i.e., diverse)

Figure 9: Results for the query "accordion player": the original ranking (top) and the results-list that has been re-ranked based on intent classes (bottom).

interpretations of the query. By diversifying the pool of possible results, one can increase the likelihood of providing the user with the information needed [8]. There are two reasons why we believe intentional framing could be potentially helpful: 1. Intentional framing naturally reflects aspects of photos that are expressed by photographers. 2. Intentional framing could be processed in a direct, light-weight fashion without extensive computational resources. In view of this motivation, we conducted an experiment on intent-based search result diversification.

6.1 Data set and set up

We evaluate the ability of our intent-based approach to improve the diversity of image search results using the task definition, development data set, and ground truth that were released by the MediaEval 2016 Retrieving Diverse Social Images Task [7]. The development data set consists of 110 general-purpose, multi-topic queries defined by the task organizers and 33,000 Creative Commons licensed images collected from Flickr.

Additionally, all images are labeled by expert annotators in terms of both relevance and diversity. For each query and its corresponding images, the relevance is judged by aggregating results from yes/no annotations. To create the ground truth for image search result diversity, images are grouped into clusters based on visual similarity. For example, images associated with query term *accordion_player* are grouped into 18 clusters including *accordion player on boat*, *accordion player standing street art*. Note that only relevant images are annotated for diversity.

Diversification performance is evaluated in terms of Cluster Recall at 5, 10 and 20 (CR@X), a measure that assesses how many different clusters from the ground truth are represented among the top X results, thus, it reflects the diversification quality of a given image result set.

We compared our intent-based diversification approach with the Flickr baseline [9]. We also compare our approach with a state-of-the-art diversification approach, designated "UPMC@MediaEval16", which is chosen because it was the top-performing approach on the MediaEval Retrieving Diverse Social Images Task in 2016 [22]. UPMC@MediaEval16 adopts a four-step approach. The first step is to create a refined initial ranked list by re-ranking the Flickr baseline using textual features (vector space model with tf-idf weights) with the aim of increasing precision. After that, the top N images are grouped using Hierarchical Clustering and the CNN features released with the data set. For each of the resulting sub-clusters, the images are re-sorted based on their position in the refined initial ranked list. Finally, the results are re-ranked by alternating images from different clusters.

6.2 Intent-based Reranking

Our intent-based re-ranking approach is illustrated in Figure 10. It adopts the same general strategy of clustering and alternating images from clusters used by UPMC@MediaEval16. However, the difference is that instead of creating clusters by applying a clustering method to the initial ranked list, we create clusters by classifying the images in intent classes and grouping together images that belong to the same class. For each query and its ranked results list returned by Flickr, we apply a three step approach to diversify the search results. The first step in this process is to predict the intent classes of retrieved image list (i.e., Flickr baseline). Once the predictions are made, we cluster the first N results based on intent class. In our case, N equals to 20. Following this step, we pick the top-one photo without replacement for each cluster, under the assumption that new clusters reflect diversity as captured by intent. Finally, these new clusters are sorted internally based on the original rank position of the images and concatenated to create the final re-ranking list. Figure 9 presents a comparison of original retrieved result and diversified retrieved result.

Note that the main difference between our diversification approach and UPMC@MediaEval16 is that we use intent classification. However, another difference is that we do not refine the initial ranking using text-based features. The omission of such an initial re-ranking step in our approach is not critical in the context of this paper, which focuses on diversification rather than on precision.

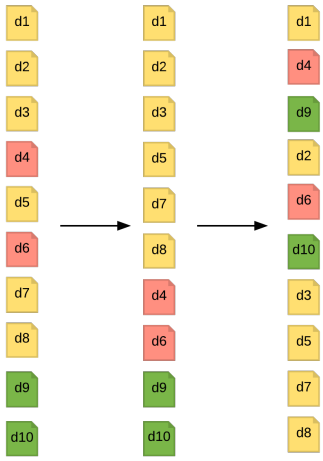

Figure 10: Re-ranking search result based on clusters of intents, colors of documents reflects different predicted intents. From left to right: initial ranking result, intent clusters, re-ranked result.

6.3 Result

Table 4 reports the results in terms of the official MediaEval 2016 evaluation metric CR@20.

Approach	CR@20	P@20
Flickr Baseline	36.1%	56.2%
UPMC@MediaEval16	49.4%	69.6%
Intent-based diversification	49.4%	56.2%

Table 4: Performance comparison against MediaEval 2016 retrieve diverse social image task over 110 queries. CR@5 and CR@10 of our approach are 27.9%, 37.5% respectively.

A significant increase can be found in terms diversity between the original Flickr baseline and UPMC@MediaEval. The most interesting aspect of this table the fact that the intent-based approach is able to achieve the same level of diversity as the state-of-the art UPMC@MediaEval16 approach. We take this result as a confirmation of the ability of classifiers trained to recognize intent to improve image retrieval algorithms. In Table 4, we also report the precision (P@20) of the three approaches. The precision gives us insight into the degree to which our approach could possibly improve if we refined the initial list with text-based features as UPMC@MediaEval16 does. Improvements in precision feed improvements in diversity, since the ground truth with respect to which cluster recall is calculated includes relevant images only.

We point out three other aspects of the intent-based diversification approach, which make it possibly more useful compared with UPMC@MediaEval16 in practice. These aspects arise because, in contrast to unsupervised learning (clustering), our approach directly tries to predict pre-defined intent classes. We mention these aspects since they represent advantages of our approach over an unsupervised learning approach, even though the two achieve the same performance with respect to the diversity metric. First, intent-based diversification has the advantage of better understandability since the classification result is able to directly provide a user-interpretable indication of the reason behind the ranking. The retrieval system can provide the user with an explanation for its prioritization of search results. Second, once the model has been trained, we do not necessarily need to fine-tune the hyper parameters, i.e., the position to cut the dendrogram. Third, image labels are generated off-line at indexing time, and a clustering step at query time, which increases the system response time, is not necessary.

7 CONCLUSION AND OUTLOOK

In this paper, we have introduced a taxonomy of intent for social photos. Additionally, we have confirmed that the visual content of images are connected to human perceptions of photographer's intent by employing a transfer learning scheme for intent classification. Moreover, we have shown that intent-based search result diversification can achieve on performance on par with that of state-of-the-art image search result diversification algorithms on a social image retrieval benchmark. Overall, this study strengthens the idea of indexing images with respect to dimensions of human perception could be important for various tasks. The study has contributed to enhancing our understanding of intentional framing. The potential of "why" dimension has been established, and future research can be undertaken to explore how intentional framing can further enhance image retrieval.

Here, we have used an intent taxonomy created by a single expert. In our future work, we are interested in the extent to which changes or refinements of this taxonomy impact the predictability of the intent classes, and the ability of the intent classes to improve the diversity of image retrieval results. We suspect that no single perfect taxonomy exists for all cases, but many taxonomies are probably "good enough". If this is the case, we are interested in optimizing the taxonomy to maximize both intuitiveness to users and the performance of the intent classifier.

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